

SDA Vision / Research report

animal-portrait.jpg



Assessment summary

Field	Recorded detail
Finding	Leaning synthetic
Recorded verdict	synthetic likely
Confidence	high
Indicative rating	78/100
Rating basis	median of available model ratings; not a calibrated probability
Explanation	3 models agree this is likely synthetic.
Coverage	image - 1 frame(s) analysed
Review context	Research assessment. Review alongside source information and context. When credentials are missing, origin remains unresolved by this check. Use these findings to inform a documented human review.

Provider overview

Provider	Type / status	Assessment	Rating
Claude Vision	vision / ok	synthetic likely	90/100
OpenAI Vision	vision / ok	synthetic likely	78/100
Gemini Vision	vision / ok	synthetic likely	75/100
C2PA Content Credentials	provenance / ok	inconclusive	n/a
Local forensic cues	forensic / ok	partially synthetic	47/100

Provider evidence / 1

Field	Recorded detail
Provider	Claude Vision
Model	claude-opus-5-5
Status	ok
Assessment	synthetic likely
Confidence	medium
Role in assessment	Supporting
Provider summary	synthetic likelihood 90/100; tells: ear anatomy ambiguous: folded ears appear to merge into the hat brim and head without a clear boundary, whiskers emerge from implausible origins (some from cheeks and neck, not only the muzzle pads) with inconsistent thickness and length, fur texture uniformly hyper-detailed and 'painted' across the face, with an overly regular tabby pattern, hat fits the cat's head perfectly with no deformation or visible support, typical of generated props
Evidence 1	Ear anatomy ambiguous: folded ears appear to merge into the hat brim and head without a clear boundary
Evidence 2	Whiskers emerge from implausible origins (some from cheeks and neck, not only the muzzle pads) with inconsistent thickness and length
Evidence 3	Fur texture uniformly hyper-detailed and 'painted' across the face, with an overly regular tabby pattern
Evidence 4	Hat fits the cat's head perfectly with no deformation or visible support, typical of generated props
Evidence 5	Idealized 'golden hour' bokeh and rim lighting consistent with stylized text-to-image output
Evidence 6	Eye reflections are glossy but generic and do not clearly correspond to the background light source
Evidence 7	Body and neck fur transition into a vague dark garment with an indistinct edge
Evidence 8	No compression-consistent sensor grain; smooth, noise-free tonal gradients throughout
Evidence 9	Caveat: A real photo of a cat wearing a hat is possible, and heavy retouching or upscaling can mimic generative artifacts. Metadata, C2PA/provenance data, and a reverse-image search should be checked before relying on this assessment.

Provider evidence / 2

Field	Recorded detail
Provider	OpenAI Vision
Model	gpt-6-astra
Status	ok
Assessment	synthetic likely
Confidence	medium
Role in assessment	Supporting
Provider summary	synthetic likelihood 78/100; tells: Chest fur forms unusually broad, sculpted clumps compared with the fine facial fur, Several whiskers have angular bends and uneven thickness, Highly regular facial detailing and unusually smooth, uniform iris textures, Hat brim and partly concealed ears have ambiguous contact geometry
Evidence 1	Chest fur forms unusually broad, sculpted clumps compared with the fine facial fur
Evidence 2	Several whiskers have angular bends and uneven thickness
Evidence 3	Highly regular facial detailing and unusually smooth, uniform iris textures
Evidence 4	Hat brim and partly concealed ears have ambiguous contact geometry
Evidence 5	Consistent warm rim lighting and plausible background blur provide some photographic cues
Evidence 6	Caveat: Stylized photography and sharpening can produce similar textures; the original file and provenance are needed to distinguish generation from photography or editing.

Provider evidence / 3

Field	Recorded detail
Provider	Gemini Vision
Model	gemini-3.8-flash
Status	ok
Assessment	synthetic likely
Confidence	medium
Role in assessment	Supporting
Provider summary	synthetic likelihood 75/100; tells: implausible ear anatomy under hat brim, hyper-smooth rendering of fur and felt texture, unnatural symmetry in whisker and facial layout, diffuse artificial lighting effects
Evidence 1	Implausible ear anatomy under hat brim
Evidence 2	Hyper-smooth rendering of fur and felt texture
Evidence 3	Unnatural symmetry in whisker and facial layout
Evidence 4	Diffuse artificial lighting effects
Evidence 5	Caveat: High-quality digital art or photo manipulation can occasionally mimic diffusion model artifacts.

Content Credentials / provenance

Field	Recorded detail
Provider	C2PA Content Credentials
Model	Not recorded
Status	ok
Assessment	inconclusive
Confidence	Not recorded
Role in assessment	Additional finding
Provider summary	File integrity: validated. Signer trust: unresolved under the current local policy. The recorded declarations remain available for review and are not used as a decisive provenance finding. Manifest declares AI-generated content.
Evidence 1	Credentials: embedded manifest found.
Evidence 2	Integrity: valid (SDK validation).
Evidence 3	Signer trust: untrusted under the configured verifier.
Evidence 4	Active manifest: Manifest declares AI-generated content.
Evidence 5	Claim generator (declared tool): Adobe_Firefly
Evidence 6	Declared signer: Adobe Inc.
Evidence 7	Declared source type: http://cv.iptc.org/newscodes/digitalsourcetype/trainedalgorithmicmedia
Evidence 8	Active record - declared tool: Adobe_Firefly; actions: c2pa.created.
Evidence 9	Active record - all actions recorded (signer's declaration): yes.
Evidence 10	Validation outcome: signingCredential.untrusted
Evidence 11	Validation outcome: timeStamp.validated
Evidence 12	Validation outcome: timeStamp.trusted
Evidence 13	Validation outcome: claimSignature.insideValidity
Evidence 14	Validation outcome: claimSignature.validated
Evidence 15	Validation outcome: assertion.hashedURI.match
Evidence 16	Validation outcome: assertion.dataHash.match

Field	Recorded detail
Evidence 17	Verifier: c2pa-python 0.37.12; native SDK 0.91.0; specification target 2.4; policy embedded-offline-reviewed-trust-v3.
Evidence 18	Offline embedded-manifest verification; no network trust-list, manifest or revocation refresh.
Evidence 19	No reviewed local C2PA trust snapshot installed. Provenance overrides are disabled.

Provider evidence / 5

Field	Recorded detail
Provider	Local forensic cues
Model	Not recorded
Status	ok
Assessment	partially synthetic
Confidence	low
Role in assessment	Additional finding
Provider summary	Heuristic synthetic likelihood 47/100 from metadata + pixel statistics.
Evidence 1	No camera EXIF (common in AI exports, screenshots, and re-saves).
Evidence 2	ELA mean 1.0, peak 26 (uniform high ELA can indicate generation/heavy edit).
Evidence 3	High-frequency noise std 3.58 (very low/very uniform noise is a synthesis cue).
Evidence 4	Spectral peakiness 32.99 (periodic frequency peaks can mark generated textures).

Evidence supporting this verdict

Field	Recorded detail
Claude Vision	synthetic likelihood 90/100; tells: ear anatomy ambiguous: folded ears appear to merge into the hat brim and head without a clear boundary, whiskers emerge from implausible origins (some from cheeks and neck, not only the muzzle pads) with inconsistent thickness and length, fur texture uniformly hyper-detailed and 'painted' across the face, with an overly regular tabby pattern, hat fits the cat's head perfectly with no deformation or visible support, typical of generated props
OpenAI Vision	synthetic likelihood 78/100; tells: Chest fur forms unusually broad, sculpted clumps compared with the fine facial fur, Several whiskers have angular bends and uneven thickness, Highly regular facial detailing and unusually smooth, uniform iris textures, Hat brim and partly concealed ears have ambiguous contact geometry
Gemini Vision	synthetic likelihood 75/100; tells: implausible ear anatomy under hat brim, hyper-smooth rendering of fur and felt texture, unnatural symmetry in whisker and facial layout, diffuse artificial lighting effects

Additional findings

Field	Recorded detail
Local forensic cues	Supporting observation; considered alongside the model assessments.
C2PA Content Credentials	File integrity: validated. Signer trust: unresolved under the current local policy. The recorded declarations remain available for review and are not used as a decisive provenance finding. Manifest declares AI-generated content.

How this assessment was reached

Field	Recorded detail
1. override	outcome=none; detail=No confirmed C2PA AI assertion or SynthID watermark. Origin remains unresolved by these checks.
2. model_agreement	synthetic=['Claude Vision', 'OpenAI Vision', 'Gemini Vision']; authentic=[]; uncertain=[]; forensic_supports_synthetic=False
3. verdict	value=synthetic_likely; confidence=high; reason=3 models agree this is likely synthetic.

Cite this analysis

Field	Recorded detail
Software (Harvard)	Martin, S. (2026) SDA Vision: Synthetic-media Discourse Analysis (version 0.1.0) [Computer software]. Manchester Metropolitan University. Available at: https://sdavision.io/
Software creator	Dr Sam Martin, Smart Data Research UK (UKRI) Fellow, Manchester Metropolitan University (MMU). ORCID: https://orcid.org/0000-0002-4466-8374 .
Funding acknowledgement	This work was supported by Smart Data Research UK, a UKRI investment; Grant number UKRI4010.
Analysis record	File: animal-portrait.jpg; media type: image; recorded at: 2026-09-26T22:31:02+00:00; models: claude-opus-5-5, gemini-3.8-flash, gpt-6-astra.

File and analysis record

Field	Recorded detail
File	animal-portrait.jpg
Analysis ID	e661b59d-245a-45f7-84ad-51f4402efb24
Generated	2026-09-26T22:31:02+00:00
Tool	SDA Vision 0.1.0
Models	claude-opus-5-5, gemini-3.8-flash, gpt-6-astra
Method version	pdf-preparation-2026-09-26.1
Analysis lens	general
Original SHA-256	bdc5c19d4211d9c9f9f8dce5bf4e4dec29e9b32e308ff12a60979481e928154f
Ingestion note 1	Fresh provider replies acquired for eleven-examples-refresh-20260926-r1; report assembled locally from the saved responses without further API calls.
Ingestion note 2	Provider latency values are restored from the acquisition receipts (maximum sampled-frame latency per provider); raw replies and assessment results are unchanged by this bookkeeping step.